	$= 0.61\text{ m}$
3(a)	The fields in both cases are uniform and vertical, such that the gravitational force and electric force act downwards on the object. This leads to the object moving at a constant downward acceleration by Newton's second law of motion in both cases, thus resulting in the parabolic path as shown in Fig. 3.1.
3(b)(i)	The gravitational force between the protons are attractive, while the electric force between them is repulsive. The opposite directions of the forces lead to them having opposite signs.
3(b)(ii)	The magnitude of the mass of a proton is much smaller than the magnitude of its charge. This means that the gravitational force between the protons will be much smaller than the electric force between them at any given separation. Therefore, the graphs are not drawn to the same scale.
3(c)(i)	The electric field strength due to both P and R at any point along the straight line always acts to the right and is non-zero. Hence, the resultant field strength will not be zero at any point along the line.
3(c)(ii)	The electric potential at the points along the straight line due to P and R individually is positive and negative respectively. Hence there will be a point on the line where the scalar sum of the potential adds up to zero.

3(c)(iii)	This occurs on the right of R, where the electric forces on the electron by P and R are leftwards and rightwards respectively. Let *r* be the distance of the electron from R. $F_P = F_R$ $E_P = E_R$ $\dfrac{6.5\times10^{-9}}{4\pi\varepsilon(r+0.060)^2} = \dfrac{0.45\times10^{-9}}{4\pi\varepsilon r^2}$ $\dfrac{r}{r+0.060} = \sqrt{\dfrac{0.45}{6.5}} = 0.263$ $r = 0.263(r+0.060)$ $r = 0.0214\text{ m}$ $\text{distance from P} = 0.060 + 0.0214$ $= 0.0814\text{ m}$ $= 8.14\text{ cm}$
4(a)	When the gas particles approach the walls of the vessel, they collide with the wall. The wall